MIL-H-11689C
8 November 1974
SUPERSEDING
MIL-H-11689B
23 April 1968

MILITARY SPECIFICATION

HAMMER, PNEUMATIC, PORTABLE

NAIL-DRIVING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a pneumatic nail-driving hammer.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-P-40	- Packaging and Packing of Hand Tools.

Military

MIL-C-12044	- Chests, Plywood.

STANDARDS

Military

MIL-STD-100	- Engineering Drawing Practices.

FSC 5130

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MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1474	- Noise Limits for Army Materiel.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street NW, Washington, DC 20036.)

DEPARTMENT OF LABOR

Title 29, Code of Federal Regulations, Chapter XVII, Part 1910, and Amendments.	- Occupational Safety and Health Administration Standards.

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(The code of Federal Regulations (CFR) and the Federal Register (FR) are on sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for the issuance thereof.)

3. REQUIREMENTS

3.1 Description. The hammer shall be air operated and shall consist of a handle assembly, a barrel, and a retainer with nail-driving tool. The entire unit shall not exceed 15 pounds in weight and shall not be over 20 inches in length.

3.2 First article (preproduction model). Unless otherwise specified (see 6.2), the supplier shall furnish one or more hammers, as specified (see 6.2), for examination and testing within the time frame specified (see 6.2), to prove, prior to starting production, that his production methods and choice of design detail will produce hammers that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and unless otherwise specified herein, all examinations and tests shall be conducted by the supplier subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests as specified (see 6.2).

3.3 Material. Material shall be as specified herein. Material not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.4 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. The identification of part numbers and changes thereto shall be in accordance with the item identification and part number requirements of MIL-STD-100. Common hardware and fastening devices, such as bolts, nuts, screws, and washers shall be held to the minimum number of types and sizes. All screw threads shall be in accordance with Handbook H28.

3.5 Temperature conditions. The hammer shall operate as specified herein in any ambient temperature from 0° F to 125° F.

3.6 Lubrication. All moving parts shall be provided with suitable means of lubrication. Oil shall be supplied by a separate line oiler.

3.6.1 Lubricants. The hammer shall operate as specified herein when lubricated with standard military lubricants.

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3.7 Handle assembly. The handle assembly shall consist of the handle, the throttle valve assembly, trigger or throttle control assembly, and air inlet. The handle shall be of the open, pistol-grip type with thumb-throttle control. An air inlet strainer (20 mesh or finer) with internal 1/4-inch NPT shall be attached to the air inlet in the handle.

3.8 Barrel. The barrel shall contain the valve mechanism and piston. The valve mechanism shall function to control the flow of air into the barrel to activate the piston.

3.9 Retainer and nail-driving tool. The retainer shall be attached to the barrel and shall hold the nail-driving tool in position to receive the blow of the piston. The nail-driving tools furnished shall be sufficient to drive the sizes of nails and spikes specified in 3.10.1.

3.10 Performance. The hammer shall operate as specified herein at any air pressure from 70 to 100 pounds per square inch (psi).

3.10.1 Capacity. The hammer shall be capable of driving nails from 10d to 60d, and spikes up to and including 3/8-inch diameter, into well-seasoned oak planks or timbers without bending more than 20 percent of any one size of nail or spike. Before the spikes are driven, pilot holes shall be predrilled the length of the spike and [illegible]-inch less in diameter than the spike. Both nails and spikes shall be driven flush with the surface of the wood.

3.10.2 Air consumption. Air consumption of the hammer shall not exceed 35 cubic feet of air per minute while the hammer is operating at full load at an air pressure of 90 psi at the tool.

3.11 Chest. When specified (see 6.2), a metal or plywood chest of sufficient size to contain the hammer, repair parts, and maintenance tools shall be furnished. When the chest is metal, it shall be made of sheet steel not less than 0.048 inch in nominal thickness and shall weigh not more than 18 pounds. Holding brackets shall be provided for holding the hammer in place, and the top shall be hinged and equipped with a carrying handle and hinged hasp. When the chest is plywood, it shall weigh not more than 12 pounds and shall conform to MIL-C-12044.

3.12 Identification marking. The hammer shall be identified in accordance with MIL-STD-130, except the required markings for the equipment shall be limited to the supplier's name or trademark, model, serial number, and stock number.

3.13 Treatment and painting. Unless otherwise specified (see 6.2), all exposed surfaces of the hammer shall be treated or painted in accordance with the supplier's standard practice.

3.14 Safety. Guards and other appropriate protection shall be provided for the operator(s) of the hammer in accordance with appropriate Department of Labor Occupational Safety and Health Administration Standards.

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3.15 Sound level. The sound level of the hammer shall be in accordance with MIL-STD-1474, Category C (90dB(A)) for steady state noise when tested in accordance with 4.5.2.5. The precautions of MIL-STD-1474 (see 6.6) shall be provided if the sound level exceeds 85dB(A)when tested in accordance with 4.5.2.5. Hearing protectors need not be supplied with the hammer. Hazard signs, if required, shall state: "CAUTION, HEARING PROTECTION REQUIRED".

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.7).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction hammer shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The preproduction hammer shall be tested as specified in 4.5.2.1, 4.5.2.2, 4.5.2.3, and 4.5.2.5. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each hammer shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

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4.4.2 Tests. Each hammer shall be tested as specified in 4.5.2.4. Failure of the test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. Hammers shall be surface examined as specified herein, to determine conformance to the requirements of this specification in general and specifically for the following defects:

Major

101. Parts and components missing.
102. Identification marking missing or not as specified.
103. Treatment and painting not as specified.
104. Repair parts, maintenance tools, and technical publications not furnished or not as specified, when applicable.

4.5.2 Tests.

4.5.2.1 Performance. The hammer shall be used to drive three groups each of 10d, 20d, 40d, and 60d nails and one group of 3/8-inch-diameter by 12-inch-long spikes into well-seasoned oak planks or timbers. The groups shall consist of 10 each nails or spikes. Both nails and spikes should be driven flush with the surface of the wood. The supplier's skilled operator may be used to operate the nail driver. Pilot holes 5/16 inch in diameter shall be predrilled the full length of the 3/8-inch spikes. The three tests shall be conducted at three air pressures ranging from 70 to 100 psi in increments of approximately 10 psi. Evidence of one or more of the following shall constitute failure of this test:

(a) More than two nails or spikes in any single group are bent during driving.
(b) Retainer does not retain the driver when the trigger is held open for 15 seconds in free air.
(c) Hammer fails to operate at any air pressure between 70 to 100 psi.

4.5.2.2 Air consumption. An air consumption test shall be conducted to determine conformance to 3.10.2. A flowmeter of common type shall be used to measure airflow. An air pressure of 90 psi shall be maintained at the tool during tests and shall be checked with a hypodermic-type gage. Air consumption shall be measured for four 30-second periods while nails or spikes are driven at full throttle. Average air consumption for the four periods exceeding 35 cubic feet per minute shall constitute failure of this test.

4.5.2.3 Climatic. The hammer shall be tested at 0° F to determine its ability to operate without freezing in low ambient temperatures. If at

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the time of testing weather conditions are not such as to provide a 0° F ambient temperature, test conditions shall be artificially produced by means of scientifically controlled and approved processes. If artifically produced test conditions are necessary, a relative humidity of 100 percent shall be maintained for the 0° F temperature. The hammer shall be thoroughly cleaned and lubricated with a commercial Grade No. 10 motor oil and shall then be allowed to cold-soak in the 0° F temperature for at least 12 hours prior to testing. The hammer shall then be used in the 0° F temperature to drive nails and spikes for 30 minutes. Evidence of the hammer freezing up or failure of the hammer to operate as specified herein during this period shall constitute failure of this test. The test shall be repeated at 125° F and prevailing weather conditions.

4.5.2.4 Operation. Each hammer shall be connected to an air supply line and operated for 20 seconds to determine correctness of assembly and ability to operate. Inability to operate as specified herein shall constitute failure of this test.

4.5.2.5 Noise level measurement.

4.5.2.5.1 Operator's station. Measure the noise at the operator's station in accordance with MIL-STD-1474 for steady-state noise while operating in accordance with 4.5.2.4. Nonconformance to 3.15 shall constitute failure of this test.

4.5.2.5.2 Hazard precautions. Measure the noise level at four positions around the hammer while operating in accordance with 4.5.2.1. Should the noise level exceed 85bB(A) at any position, failure to provide the precautions specified in 3.15 shall constitute failure of this test.

4.7 Inspection of preparation for delivery.

4.7.1 Quality conformance inspection of pack.

4.7.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.7.1.2 Examination. The preservation, packaging, and marking shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

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105. Materials, methods, and containers not as specified. Each incorrect material, method, or container shall be considered one defect.
106. Preservation not in accordance with the referenced document for Level A.
107. Consolidated packaging not as specified for Level A.
108. Strapping not as specified.
109. Marking missing, illegible, incorrect or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Preservation. The hammer, repair parts, maintenance tools, and technical publications shall be preserved in accordance with PPP-P-40, Level A.

5.1.1.2 Consolidated packaging. When a chest is furnished (see 3.11), the hammer, repair parts, maintenance tools, and technical publications shall be consolidated packaged in the chest. The chest lid shall be closed and secured. When a chest is not furnished, the hammer, repair parts, maintenance tools, and technical publications shall be consolidated packaged in a box conforming to PPP-B-636, Class Weather Resistant. The contents shall be blocked and cushioned to prevent damage and free movement. The box shall be closed, Method V, in accordance with the box specification appendix.

5.1.2 Level C. Each complete hammer with repair parts, tools, and publications shall be preserved and packaged to afford protection against deterioration and damage from the supplier to the initial destination. Chests, when furnished (see 3.11), shall be utilized as the packaging container. The supplier's standard commercial preservation and packaging practice is acceptable when it fulfills these requirements.

5.2 Packing. Packing shall be Level A, B, or C as specified (see 6.2).

5.2.1 Level A. Hammers, preserved and packaged as specified in 5.1 shall be packed in a close-fitting box conforming to PPP-B-601, Overseas Type, Grade B, style optional. The box shall be closed and strapped in accordance with the appendix to the box specification. Strapping shall be in accordance with the appendix to the box specification.

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5.2.2 Level B. Hammers, preserved and packaged as specified in 5.1 shall be packed in a close-fitting box conforming to PPP-B-601, Domestic Type, Grade B, style optional or PPP-B-640, Class 2, style optional. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.3 Level C. Hammers, preserved and packaged as specified in 5.1 shall be packed in a manner to assure carrier acceptance and safe delivery to destination at lowest rates in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. Packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The hammer is intended for use in driving nails.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When preproduction model is not required (see 3.2).
(c) Time frame required for submission of preproduction model and number of preproduction hammers required (see 3.2).
(d) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the supplier (see 3.2).
(e) When seal or metal or plywood chest is required (see 3.11).
(f) Treatment and painting required when other than as specified (see 3.13).
(g) Level of preservation and packaging and level of packing required (see 5.1 and 5.2).

6.3 Preproduction model. Any changes or deviations of production hammers form the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish hammers conforming to this specification.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and suppliers maintenance and operation manual(s) to be furnished with each hammer.

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6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of hammers.

6.6 Noise limits. When the noise level of the hammer exceeds 85dB(A), the appropriate discussion of the noise hazard in accordance with MIL-STD-1474 (MANUAL Paragraph) shall include the requirement for hearing protection, the noise level of the hammer, and the distance from the hammer at which 85dB(A) will always be met.

Custodians:

Army - ME
Air Force - 84

User activities:

Army - GL
Navy - YD

Preparing activity:

Army - ME

Project No. 5130-0217

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INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE MIL-H-[illegible], Hammer, Pneumatic, Portable Nail-Driving

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

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